
PO33Q - The Life and Death of Democracies and Dictatorships: A Quantitative Perspective

Week 8 - Solutions

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Exercise 1: From a static probit to a Markov Transition Model (Modernisation)

Model Estimation

```
probit_1 <- glm(democracy ~ l.log_gdppc + l.gdppc_growth,
               data = world, na.action = na.exclude,
               family = binomial(link = "probit"))

probit_ldv <- glm(democracy ~ l.log_gdppc + l.gdppc_growth + l.democracy,
                 data = world, na.action = na.exclude,
                 family = binomial(link = "probit"))

# Step 3: plain MTM (CRE-free baseline)
emergence_1 <- glm(democracy ~ l.log_gdppc + l.gdppc_growth,
                  data = world_democ0, na.action = na.exclude,
                  family = binomial(link = "probit"))
survival_1 <- glm(democracy ~ l.log_gdppc + l.gdppc_growth,
                  data = world_democ1, na.action = na.exclude,
                  family = binomial(link = "probit"))

# Step 3b: MTM with Correlated Random Effects (country means added)
emergence_1_cre <- glm(democracy ~ l.log_gdppc + l.gdppc_growth +
                      mean_log_gdppc + mean_gdppc_growth,
                      data = world_democ0, na.action = na.exclude,
                      family = binomial(link = "probit"))
survival_1_cre <- glm(democracy ~ l.log_gdppc + l.gdppc_growth +
                      mean_log_gdppc + mean_gdppc_growth,
                      data = world_democ1, na.action = na.exclude,
                      family = binomial(link = "probit"))
```

```

library(sandwich)
library(lmtest)

coeftest(emergence_1_cre,
         vcov. = vcovCL(emergence_1_cre, cluster = ~ countrycode, type = "HC3"))
##
## z test of coefficients:
##
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -1.8856397   0.2712365  -6.9520 3.601e-12 ***
## l.log_gdppc     0.0200140   0.0387428   0.5166   0.6054
## l.gdppc_growth -0.0016900   0.0030731  -0.5499   0.5824
## mean_log_gdppc -0.0216691   0.0604906  -0.3582   0.7202
## mean_gdppc_growth -0.0192134  0.0172461  -1.1141   0.2652
## -
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
coeftest(survival_1_cre,
         vcov. = vcovCL(survival_1_cre, cluster = ~ countrycode, type = "HC3"))
##
## z test of coefficients:
##
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -0.94167826   0.38232695  -2.4630  0.01378 *
## l.log_gdppc     0.29985416   0.07368164   4.0696 4.71e-05 ***
## l.gdppc_growth  0.00073211   0.00412205   0.1776  0.85903
## mean_log_gdppc  0.14609817   0.08114331   1.8005  0.07178 .
## mean_gdppc_growth -0.03816786  0.04299995  -0.8876  0.37474
## -
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

prob_p1 <- predict(probit_1, type = "response"); world$prob_p1 <- unlist(prob_p1)
roc_p1 <- roc(world$democracy, world$prob_p1)
prob_ldv <- predict(probit_ldv, type = "response"); world$prob_ldv <- unlist(prob_ldv)
roc_ldv <- roc(world$democracy, world$prob_ldv)
prob_em1 <- predict(emergence_1_cre, type = "response"); world_democ0$prob_em1 <- unlist(prob_em1)
roc_em1 <- roc(world_democ0$democracy, world_democ0$prob_em1)
prob_sur1 <- predict(survival_1_cre, type = "response"); world_democ1$prob_sur1 <- unlist(prob_sur1)
roc_sur1 <- roc(world_democ1$democracy, world_democ1$prob_sur1)

cat("AUC static:", round(auc(roc_p1),3), " LDV:", round(auc(roc_ldv),3),
    " emerg:", round(auc(roc_em1),3), " surv:", round(auc(roc_sur1),3), "\n")
## AUC static: 0.766 LDV: 0.987 emerg: 0.526 surv: 0.845

```

Solutions to Reflection Questions

Q1. From static to dynamic: why does income shrink when $\mathbf{1} \cdot \mathbf{democracy}$ is added, and what does $\mathbf{1} \cdot \mathbf{democracy}$ absorb?

In the static probit, $\mathbf{1} \cdot \log_gdppc$ has a coefficient of 0.388 and is highly significant – richer countries are far more likely to be democracies. Once $\mathbf{1} \cdot \mathbf{democracy}$ is added, income falls to 0.142, a reduction of roughly two thirds, though it stays significant. $\mathbf{1} \cdot \mathbf{democracy}$ absorbs the strong path dependence of regime type: a country that was a democracy last year is overwhelmingly likely to be one this year, and those countries also tend to be richer. In the static model income was partly standing in for regime durability – rich, stable democracies contribute many repeated, easily-predicted observations – so it looked more powerful than it is. Once last year's regime is controlled for, income's residual effect is much smaller. A researcher who stopped at the static probit would have read this as strong support for modernisation theory, when much of it was really a restatement of the fact that regimes persist. (Steps 1 and 2 are deliberately CRE-free baselines, so these two figures are unaffected by the country means added later.)

Q2. Why is splitting into emergence and survival the right move – what can the MTM see that the LDV cannot?

Including $\mathbf{1} \cdot \mathbf{democracy}$ as a single additive term assumes its effect is the same regardless of context: last year's regime shifts the baseline probability of democracy by a fixed amount whether the country is trying to escape autocracy or to hold on to democracy. But those are exactly the situations we expect to differ – wealth might matter for one and not the other. The LDV imposes one adjustment on both.

The MTM drops that assumption by splitting the data: emergence is estimated only on country-years that were autocratic last year, survival only on those that were democratic. Each model gets its own intercept and its own coefficients, so the effect of income on emergence and on survival are estimated completely separately, with no requirement that they match. And in this data they plainly do not: the within-country income coefficient is small and weak in the emergence model but sizeable and positive in survival. The substantive reading is the Przeworski one – it is less that wealth *creates* democracies than that wealthy democracies rarely *break down*. Emergence looks like a contingent, historically specific event only loosely tied to income; survival is the more systematic function of development. (Note that because the CRE means are now included, the survival income coefficient no longer coincides with the static probit's 0.388 – the means have stripped the between-country component out of it. GDP growth is negligible and insignificant in both, suggesting it is the level, not the rate, of development that matters.)

Q3. Why is survival so much easier to predict than emergence?

The AUCs tell the story: the static probit reaches 0.766, the LDV probit leaps to 0.987 (last year's regime is a near-perfect predictor of this year's), the survival model scores high, and the emergence model sits only marginally above the 0.5 diagonal. The gap between the two MTM models is not about model quality – it is about the prediction problem. Democratic survival is easy because breakdowns are rare: a model that simply guessed “still a democracy” for every observation would already be right most of the time. Emergence is hard because transitions are rare in the opposite direction – there are only around 105 democratic transitions in the whole dataset – so the model is trying to forecast a genuinely uncommon event. A high survival AUC should be read cautiously; a merely respectable emergence AUC is the more impressive achievement. (The emergence and survival AUCs are not on the same scale as the pooled models, since they come from different subsets with very different class balances.)

Going further: year fixed effects

```
emergence_no_fe <- glm(democracy ~ l.log_gdppc + l.gdppc_growth,
  data = world_democo, na.action = na.exclude,
  family = binomial(link = "probit"))
emergence_fe <- glm(democracy ~ l.log_gdppc + l.gdppc_growth + factor(year),
  data = world_democo, na.action = na.exclude,
  family = binomial(link = "probit"))
lrtest(emergence_no_fe, emergence_fe)
## Likelihood ratio test
##
## Model 1: democracy ~ l.log_gdppc + l.gdppc_growth
## Model 2: democracy ~ l.log_gdppc + l.gdppc_growth + factor(year)
##   #Df LogLik Df  Chisq Pr(>Chisq)
## 1    3 -464.44
## 2   62 -421.85 59 85.181    0.01448 *
## -
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

coef_df <- as.data.frame(summary(emergence_fe)$coefficients)
coef_df <- coef_df[grepl("factor\\(year\\)", rownames(coef_df)), ]
head(coef_df[order(-coef_df$`Std. Error`), ], 5)
##               Estimate Std. Error      z value Pr(>|z|)
## factor(year)1962 0.08505220   2562.768 3.318762e-05 0.9999735
## factor(year)1964 0.09089358   2554.744 3.557835e-05 0.9999716
## factor(year)1967 0.09363052   2551.884 3.669074e-05 0.9999707
## factor(year)1968 0.09347959   2550.225 3.665543e-05 0.9999708
## factor(year)1969 0.09576476   2549.069 3.756852e-05 0.9999700
```

```
survival_no_fe <- glm(democracy ~ l.log_gdppc + l.gdppc_growth,
  data = world_democ1, na.action = na.exclude,
  family = binomial(link = "probit"))
survival_fe <- glm(democracy ~ l.log_gdppc + l.gdppc_growth + factor(year),
  data = world_democ1, na.action = na.exclude,
  family = binomial(link = "probit"))
lrtest(survival_no_fe, survival_fe)
## Likelihood ratio test
##
## Model 1: democracy ~ l.log_gdppc + l.gdppc_growth
## Model 2: democracy ~ l.log_gdppc + l.gdppc_growth + factor(year)
##   #Df LogLik Df  Chisq Pr(>Chisq)
## 1    3 -301.26
## 2   62 -269.35 59 63.814    0.3111

coef_df_sur <- as.data.frame(summary(survival_fe)$coefficients)
coef_df_sur <- coef_df_sur[grepl("factor\\(year\\)", rownames(coef_df_sur)), ]
head(coef_df_sur[order(-coef_df_sur$`Std. Error`), ], 5)
##               Estimate Std. Error      z value Pr(>|z|)
## factor(year)1971 0.4809058   4131.793 1.163915e-04 0.9999071
## factor(year)1974 0.5138300   4125.159 1.245600e-04 0.9999006
## factor(year)1975 0.3991099   4123.434 9.679067e-05 0.9999228
## factor(year)1978 0.1960084   4123.020 4.754001e-05 0.9999621
## factor(year)1982 0.3414285   4115.280 8.296605e-05 0.9999338
```

The likelihood-ratio test gives a clear answer that differs across the two models, and the lesson is in the gap between statistical significance and practical feasibility.

- For **emergence** the test is significant (LR = 85.18, df = 59, $p = 0.015$), so there is detectable across-year variation in the baseline transition probability – but the year dummies are not usable: the worst-affected years carry standard errors of around 2,500 (e.g. `factor(year)1962` has SE 2,563 on a point estimate of 0.085). That is the signature of near-complete separation: in years with no transitions, the dummy almost perfectly predicts the outcome and cannot be estimated reliably even when the point estimate looks finite.
- For **survival** the test is insignificant (LR = 63.81, df = 59, $p = 0.311$) *and* separation is worse still – standard errors above 4,100 (e.g. `factor(year)1971`, SE 4,132), because those are years in which every democracy survived. The conclusion is to include year fixed effects in neither model: in emergence they are significant but unreliable, in survival neither significant nor feasible. This is a recognised limitation of the MTM on rare binary outcomes; the standard response in the literature is to omit them and rely on country-clustered standard errors, and to say so.

Exercise 2: Predict, then check (Dependency)

Model Estimation

```
probit_2 <- glm(democracy ~ l.agri + l.natural + l.log_gdppc,
               data = world, na.action = na.exclude,
               family = binomial(link = "probit"))

emergence_2_cre <- glm(democracy ~ l.agri + l.natural + l.log_gdppc +
                      mean_agri + mean_natural + mean_log_gdppc,
                      data = world_democo, na.action = na.exclude,
                      family = binomial(link = "probit"))
survival_2_cre <- glm(democracy ~ l.agri + l.natural + l.log_gdppc +
                     mean_agri + mean_natural + mean_log_gdppc,
                     data = world_democ1, na.action = na.exclude,
                     family = binomial(link = "probit"))

coeftest(emergence_2_cre, vcov. = vcovCL(emergence_2_cre, cluster = ~ countrycode, type = "HC3"))
##
## z test of coefficients:
##
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -2.97711043  0.87629866 -3.3974 0.0006804 ***
## l.agri        0.00072626  0.01046981  0.0694 0.9446974
## l.natural     0.01822510  0.01091445  1.6698 0.0949563 .
## l.log_gdppc   -0.01116170  0.08370748 -0.1333 0.8939231
## mean_agri     0.01432397  0.01370461  1.0452 0.2959335
## mean_natural  -0.05432474  0.01313650 -4.1354 3.543e-05 ***
## mean_log_gdppc 0.15223995  0.12675076  1.2011 0.2297136
## —
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
coeftest(survival_2_cre, vcov. = vcovCL(survival_2_cre, cluster = ~ countrycode, type = "HC3"))
##
## z test of coefficients:
##
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -0.0548707  0.9838730 -0.0558 0.9555
## l.agri        0.0022676  0.0168871  0.1343 0.8932
## l.natural     0.0169110  0.0177963  0.9503 0.3420
## l.log_gdppc   0.2141998  0.1447472  1.4798 0.1389
## mean_agri     -0.0111142  0.0181426 -0.6126 0.5401
## mean_natural  -0.0136805  0.0183053 -0.7474 0.4549
## mean_log_gdppc 0.1029721  0.1921349  0.5359 0.5920
```

Solutions to Reflection Questions

Q1. The prediction most students get wrong: where does the resource curse actually show up?

The natural prediction from dependency theory is that $\ell.natural$ is negative and significant – a country growing more resource-dependent becomes less democratic. The data do not bear that out for the *lagged within-country* term. In the emergence model the significant negative resource term is the country mean, $mean_natural = -0.054$ ($p < 0.001$); the within-country $\ell.natural$ is small, positive and marginal ($+0.018$, $p \approx 0.095$), and in survival no resource term comes close ($\ell.natural +0.017$, $p \approx 0.34$). Substantively: persistently resource-dependent countries are less likely to make democratic transitions, but a country becoming more resource-dependent *than its own historical norm* is not measurably less likely to. The curse here is a cross-country regularity, not a year-to-year dynamic – which fits the idea that rents fund durable repressive and patronage structures that distinguish countries over the long run rather than switching democratic prospects on and off annually. (In the raw static probit $\ell.natural$ is negative, as expected, and $\ell.agri$ is actually positive once income and rents are controlled for – the same surprise seen in the OLS exercises; every $\ell.agri$ term, lagged or mean, is insignificant in the CRE models.) Because the signal sits on a Mundlak mean we treat as a nuisance term, the honest reading is that the cleaner way to study the curse head-on would be a between-effects or pooled model, not the within-country MTM.

Q2. Does income still carry survival once commodity dependence is in the model?

No – and that is the second thing worth noticing. In Exercise 1 the income–survival link was the clean, robust result. Here, with agriculture and natural resources (and their means) in the specification, $\ell.log_gdppc$ in the survival model is 0.214 with $p \approx 0.14$ – no longer significant. The Exercise 1 finding was real but not bulletproof: it weakens once correlated structural controls enter. This is a small lesson in humility about headline results, and a reason to report the specification a coefficient lives in, not just the coefficient.

Q3. How cautiously should the emergence coefficients be read?

Carefully. The emergence model uses only the subset of country-years that were autocratic the year before, and the event it is modelling – an actual transition – is rare, roughly 105 in the full data. The coefficients are therefore identified from a small number of real events, the standard errors are large relative to the estimates, and an insignificant variable may still have a real effect the model simply lacks the power to detect. Adding the CRE means spends further degrees of freedom on this already-thin subset, which is one more reason to read the emergence estimates conservatively.

Exercise 3: Diagnose the model (Resource Curse)

Model Estimation

```
probit_3 <- glm(democracy ~ l.oil + l.v2x_neopat + l.log_gdppc,
               data = world, na.action = na.exclude,
               family = binomial(link = "probit"))

emergence_3_cre <- glm(democracy ~ l.oil + l.v2x_neopat + l.log_gdppc +
                      mean_oil + mean_v2x_neopat + mean_log_gdppc,
                      data = world_democo, na.action = na.exclude,
                      family = binomial(link = "probit"))
survival_3_cre <- glm(democracy ~ l.oil + l.v2x_neopat + l.log_gdppc +
                     mean_oil + mean_v2x_neopat + mean_log_gdppc,
                     data = world_democ1, na.action = na.exclude,
                     family = binomial(link = "probit"))

coeftest(emergence_3_cre, vcov. = vcovCL(emergence_3_cre, cluster = ~ countrycode, type = "HC3"))
##
## z test of coefficients:
##
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -1.0443218   0.6031149  -1.7315   0.08335 .
## l.oil          -0.0108249   0.0091949  -1.1773   0.23909
## l.v2x_neopat   -0.8779776   0.5725138  -1.5335   0.12514
## l.log_gdppc    -0.0438063   0.0641845  -0.6825   0.49492
## mean_oil       -0.0260272   0.0125951  -2.0665   0.03879 *
## mean_v2x_neopat -0.3781136   0.5702346  -0.6631   0.50728
## mean_log_gdppc  0.0467490   0.1000339   0.4673   0.64026
## -
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
coeftest(survival_3_cre, vcov. = vcovCL(survival_3_cre, cluster = ~ countrycode, type = "HC3"))
##
## z test of coefficients:
##
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    0.801300   0.924074   0.8671   0.38587
## l.oil           0.031970   0.048936   0.6533   0.51356
## l.v2x_neopat   -1.812175   0.865595  -2.0936   0.03630 *
## l.log_gdppc     0.172731   0.098558   1.7526   0.07967 .
## mean_oil       -0.047741   0.039211  -1.2175   0.22340
## mean_v2x_neopat  0.327647   0.991892   0.3303   0.74115
## mean_log_gdppc  0.095746   0.139041   0.6886   0.49106
## -
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Solutions to Reflection Questions

Q1. Trap one: is the oil curse a within-country dynamic, or a fact about which countries have oil?

Oil is the starkest between-country variable in the whole set – a handful of states hold most of the rents, most hold almost none – so `mean_oil` does most of the work. Its only significant term anywhere is the between-country mean in the emergence model (`mean_oil` = -0.026, $p \approx 0.039$): persistently oil-rich countries are less likely to democratise. The within-country lagged term is not significant in either model (`l.oil` = -0.011, $p \approx 0.24$ in emergence; +0.032, $p \approx 0.51$ in survival). So the oil-autocracy link here is a cross-country regularity, not a within-country process – a country pumping more oil than its own historical average does not thereby become measurably less democratic. (In the raw static probit `l.oil` is negative and significant, and it shrinks once `l.democracy` enters, for the familiar reason: oil-rich autocracies have long been autocratic, and the lagged regime absorbs that persistence.) The careless reading – “oil causes autocracy, look at the significant oil term” – mistakes a between-country mean, formally a nuisance term, for a dynamic effect.

Q2. Trap two: the thing that actually threatens democracy here is not oil.

The standard story expects oil and neopatrimonialism to block emergence. The data say something close to the opposite. The term that reaches significance is neopatrimonialism in the survival model: `l.v2x_neopat` = -1.81 ($p \approx 0.036$). A significant negative coefficient here is consistent with backsliding – established democracies drifting toward personalised, clientelistic rule and eventually slipping back below the democratic threshold. Read it cautiously, though. `v2x_neopat` is itself built partly from measures of executive constraint, judicial independence and corruption, so it captures the texture of how power is exercised rather than something cleanly separable from democratic quality – and the causal arrow plausibly runs both ways (weak democracies let clientelism flourish, not only the reverse). The overlap is milder here than it would be against a measure like Polity, because the dependent variable is the minimalist Boix-Miller-Rosato threshold, which scores only elections and suffrage; but it is not zero. In the emergence model the same term is negative but clearly insignificant (-0.88, $p \approx 0.13$), and the between-country means (`mean_v2x_neopat`) are insignificant in both. The safe reading is that neopatrimonialism is a survival-side correlate – the action is on survival, not emergence – rather than a demonstrated cause of breakdown, and worth stating plainly because the asymmetry itself cuts against the familiar elite-resistance framing.

Q3. Trap three: income's survival effect does not survive this specification either.

In Exercise 1 income clearly carried survival; here, with oil and neopatrimonialism in the model, `l.log_gdppc` in survival is 0.173 with $p \approx 0.08$ – marginal at best – and in emergence it is insignificant ($p \approx 0.49$). Read alongside Exercise 2, where commodity controls also knocked income out of the survival column, the lesson is general: a headline coefficient is only ever a statement about one specification. Income looks like a robust protector of democracy until you put it next to correlated structural variables, at which point much of its apparent effect redistributes. The discipline is to report what is in the model whenever you report a coefficient, and to be wary of reading any single estimate in isolation. The clean one-line summary of the exercise: the oil curse here works against emergence and only as a cross-country regularity, while what tracks survival is neopatrimonialism, not oil.

Exercise 4: Report it (Results Table)

Producing the Table

```
library(modelsummary)
library(tinytable)
library(sandwich)
library(tibble)

vcov_list <- list(
  "cluster-robust" = vcovCL(probit_1,      cluster = ~ countrycode, type = "HC3"),
  "cluster-robust" = vcovCL(probit_ldv,    cluster = ~ countrycode, type = "HC3"),
  "cluster-robust" = vcovCL(emergence_1_cre, cluster = ~ countrycode, type = "HC3"),
  "cluster-robust" = vcovCL(survival_1_cre, cluster = ~ countrycode, type = "HC3")
)

models_ex1 <- list(
  "Static"      = probit_1,
  "LDV"         = probit_ldv,
  "Emergence"   = emergence_1_cre,
  "Survival"    = survival_1_cre
)

cm <- c(
  "l.log_gdppc"   = "Log GDP pc (lagged)",
  "l.gdppc_growth" = "GDP growth (lagged)",
  "l.democracy"   = "Democracy (lagged)",
  "(Intercept)"   = "(Intercept)"
)

rows <- tibble(
  `~term` = "AUC",
  `~(1)`  = as.numeric(auc(roc_p1)),
  `~(2)`  = as.numeric(auc(roc_ldv)),
  `~(3)`  = as.numeric(auc(roc_em1)),
  `~(4)`  = as.numeric(auc(roc_sur1))
)

attr(rows, "position") <- 9
```

```

modelsummary(
  models_ex1,
  vcov      = vcov_list,
  estimate  = "{estimate}{stars}",
  statistic = "{std.error}",
  gof_omit  = "DF|Deviance|Log.Lik|F|AIC|BIC|RMSE",
  coef_map  = cm,
  add_rows  = rows,
  notes     = c(
    "+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001",
    "Standard errors are cluster-robust (HC3), clustered by country.",
    "Emergence and Survival columns include country means (Correlated Random Effects), omitted from the
    ↪ table; Static and LDV do not."
  )
) |>
group_tt(j = list("MTM" = 4:5)) |>
group_tt(j = list("Dependent Variable: Democracy" = 2:5))|>
theme_latex(resize_width= 0.75, resize_direction="both")

```

	Dependent Variable: Democracy			
	Static	LDV	MTM	
			Emergence	Survival
Log GDP pc (lagged)	0.388*** (0.049)	0.142*** (0.024)	0.020 (0.039)	0.300*** (0.074)
GDP growth (lagged)	-0.001 (0.001)	-0.001 (0.002)	-0.002 (0.003)	0.001 (0.004)
Democracy (lagged)		4.075*** (0.105)		
(Intercept)	-2.932*** (0.344)	-3.021*** (0.165)	-1.886*** (0.271)	-0.942* (0.382)
AUC	0.766	0.987	0.526	0.845
Num.Obs.	9027	9026	4522	4504
Std.Errors	Cluster-Robust	Cluster-Robust	Cluster-Robust	Cluster-Robust

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Emergence and Survival columns include country means (Correlated Random Effects), omitted from the table; Static and LDV do not.

Solutions to Reflection Questions

Q1. How does $\ln \log_gdppc$ move across the four columns?

In a revealing pattern. The static probit gives 0.388, positive and highly significant – the textbook modernisation result. The LDV probit cuts it to 0.142, still significant but less than half, as regime persistence absorbs much of what income was doing. The two MTM columns then split the picture: the within-country income coefficient is small and weak for emergence and sizeable and positive for survival. Two things change at once as you move into the MTM columns – the conditioning on last year's regime *and* the addition of the country means – so the survival coefficient no longer matches the static 0.388, because the means have removed its between-country part. The progression as a whole shows how path dependence and unobserved heterogeneity together structure the data in a way no single pooled model reveals: income's apparent power in column 1 is partly persistence (column 2) and partly between-country difference (the CRE means in columns 3–4).

Q2. AUC: which MTM model fits better, and why is that less impressive than it looks?

The survival model almost always posts the higher AUC, often by a wide margin – but that reflects the prediction problem, not the specification. Breakdowns are rare, so the survival subset is overwhelmingly democracy = 1, and a model that guessed “democracy” everywhere would already score well. The emergence model has the harder task of forecasting rare transitions (about 105 in the data), so a respectable emergence AUC is the genuinely impressive number and a high survival AUC should be read with the base rate in mind. (The two are not comparable with the static and LDV AUCs, which are computed on the full pooled sample.)

Q3. Why is the single $\ln \log_gdppc$ democracy coefficient a poor approximation?

It forces an implausible symmetry: it assumes the effect of last year's regime on this year's is the same whether the country starts as an autocracy or a democracy. Everything in Exercises 1–3 says those are different situations – income protects survival but does little for emergence; the resource curse weighs on emergence (between countries) while backsliding threatens survival. The LDV averages over both subsets at once, producing a coefficient that describes neither process accurately and, worse, hides the central substantive finding that emergence and survival answer to different variables. The table makes the case visually: the Emergence and Survival columns disagree precisely where column 2 pretends there is nothing to disagree about.

Solutions to Synthesis Questions

Q1. Do the predictors of emergence and survival overlap, or differ?

They are related but not identical, and the CRE layer adds a second cut – within-country versus between-country – that matters here. Income is directionally a stronger predictor of survival than of emergence in all three exercises, but its significance is clean only in the modernisation model (Exercise 1, survival $p < 0.001$); once commodity dependence or oil and neopatrimonialism enter, the income–survival link weakens to marginal (Exercise 3, $p \approx 0.08$) or insignificant (Exercise 2, $p \approx 0.14$). Natural resources and oil are associated with preventing *emergence*, not undermining survival – but in both cases the effect sits on the *between-country* mean ($\text{mean_natural} = -0.054$; $\text{mean_oil} = -0.026$), not the lagged within-country term, so the resource curse shows up as a cross-country regularity rather than a year-to-year dynamic. Neopatrimonialism, by contrast, is a survival-side correlate: significant and negative in the survival model of Exercise 3 (-1.81) but not in emergence – the reverse of the elite-resistance-to-transition emphasis one might expect, though the coefficient is best read as an association given how much the index shares with democratic quality itself. The MTM's contribution is to make the emergence/survival distinction visible; the CRE layer adds the further lesson that some of what looks like a within-country effect is really a stable between-country difference, which is exactly why several headline resource-curse results turn out to be between-country findings.

Q2. The key difference between the LDV probit and the MTM.

The LDV probit includes last year's regime as one more covariate, implicitly assuming the effects of income, oil or commodity dependence are the same regardless of the previous year's regime. The MTM rejects that and estimates two entirely separate models – one for the autocracy-to-democracy transition, one for the persistence of democracy – letting every coefficient differ freely between them. It is the more appropriate tool for regime change because theory gives strong reasons to expect emergence and survival to work differently, and collapsing them into a single pooled coefficient buys simplicity at the cost of the substantive question.

Q3. What is gained and lost by treating democracy as binary?

A binary makes a clean theoretical commitment – democracy as a threshold, not a matter of degree – and lets us study transitions as the discrete events most democratisation theory is actually about. What is lost is everything happening *within* the democratic category: a consolidated liberal democracy and a fragile electoral one are both coded 1, though they may behave very differently. Survival is the more distorted of the two by this coding: the survival model asks only whether a country stays above the line, and so misses exactly the gradual erosion of norms and institutions that defines contemporary backsliding until it crosses a threshold. Emergence is by definition a binary event – a country either crosses or it does not – so the coding is more natural there.

Q4. CRE versus clustering: what does each solve, and why both?

They fix two different problems that share one source – stable, unobserved country differences. The country means (CRE, Step 3b) fix a *bias in the coefficients*: without them, a lagged predictor soaks up the influence of time-invariant traits (geography, deep institutions, political culture) correlated with both predictor and outcome, conflating within-country change with fixed between-country difference. Adding the means partials that out, leaving a cleaner within-country estimate. Cluster-robust standard errors (Step 4) fix the *uncertainty*: even with unbiased estimates, a country's observations are serially correlated, so default standard errors understate the true variability and overstate significance. CRE changes the estimates but not their standard errors; clustering changes the standard errors but not

the estimates. Correct estimates with the wrong uncertainty, or honest uncertainty around biased estimates, are each only half an answer – which is why the MTM workflow uses both.